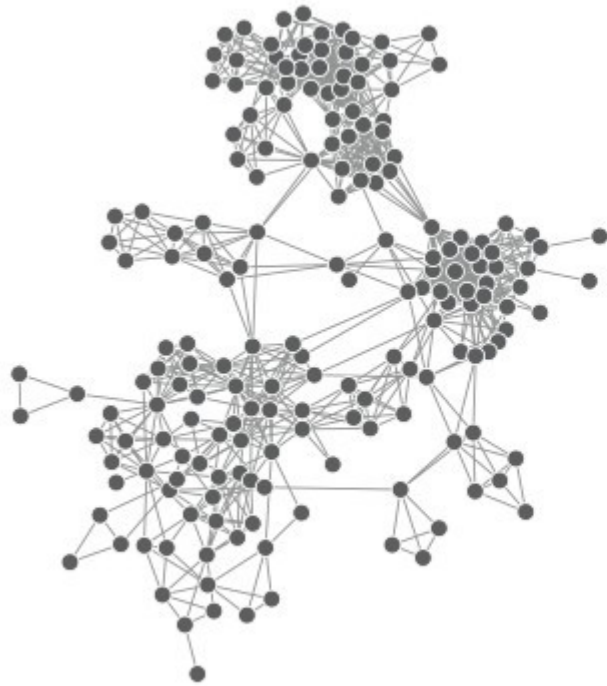


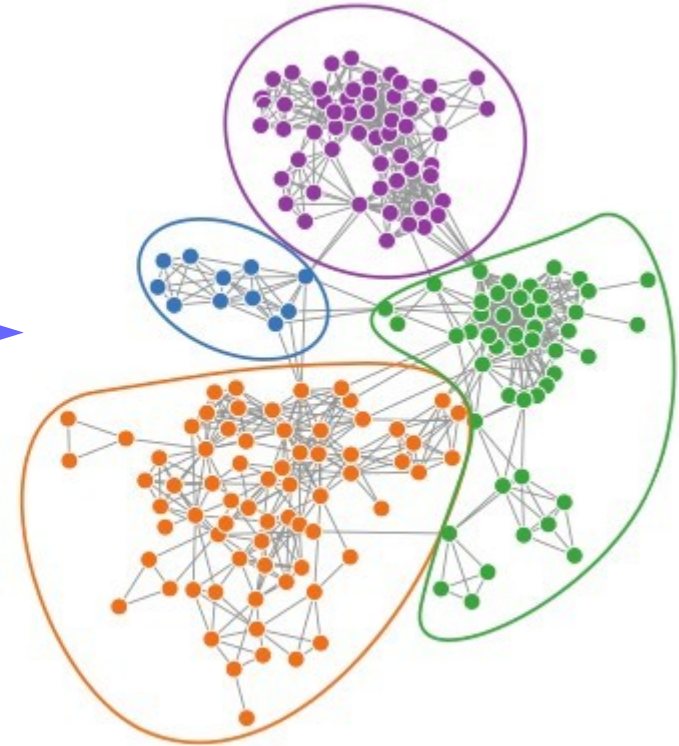
Community detection

Leto Peel
l.peel@maastrichtuniversity.nl
@PiratePeel

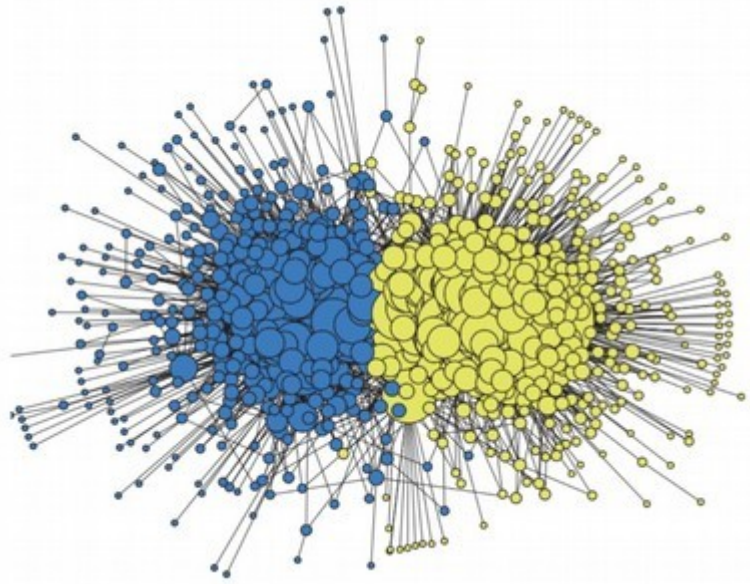
Community detection



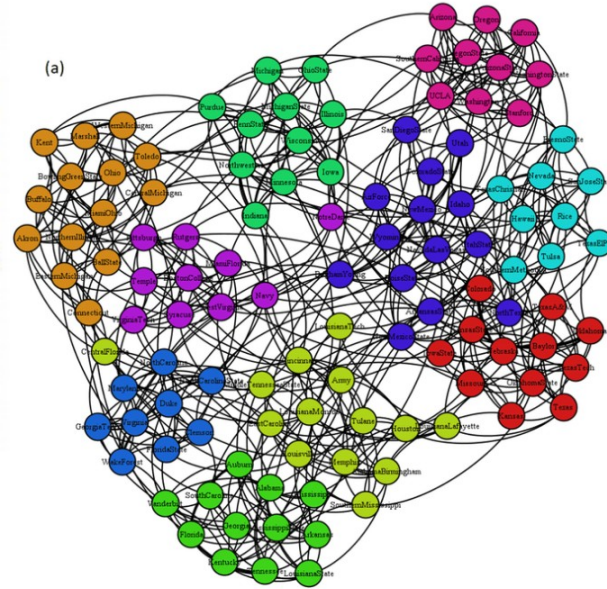
Network



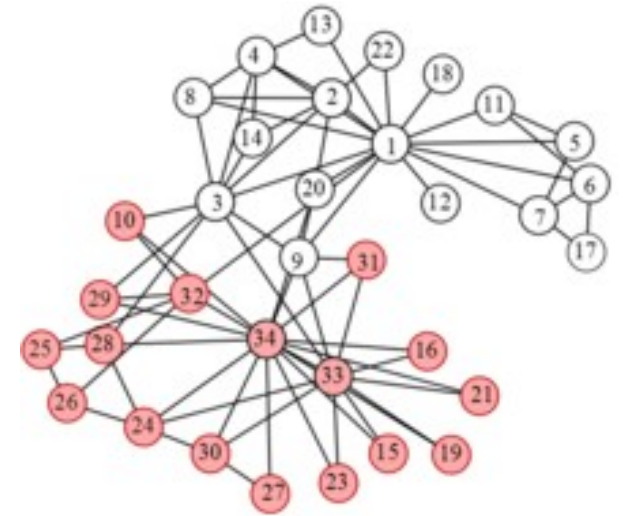
Communities



US political blogs



American football



karate club*

*this network is so popular for community detection it has its own prize associated with it, see <http://networkkarate.tumblr.com/>

Recap: the null-model ladder

Rung	Null model	What it holds fixed	Failing it tells you...
1	Erdős–Rényi	number of nodes & edges	"not maximally random" (almost everything fails this)
2	Configuration model	every node's degree	there is structure beyond degree heterogeneity
3	DC-SBM	degrees and group structure	structure beyond blocks + degrees → Wednesday's session
—	BESTest	a node attribute you have	this attribute is (not) associated with the structure

Given a partition, which one should you trust?

Supervised vs unsupervised

Supervised vs unsupervised

The supervised learner doesn't know much



Supervised vs unsupervised

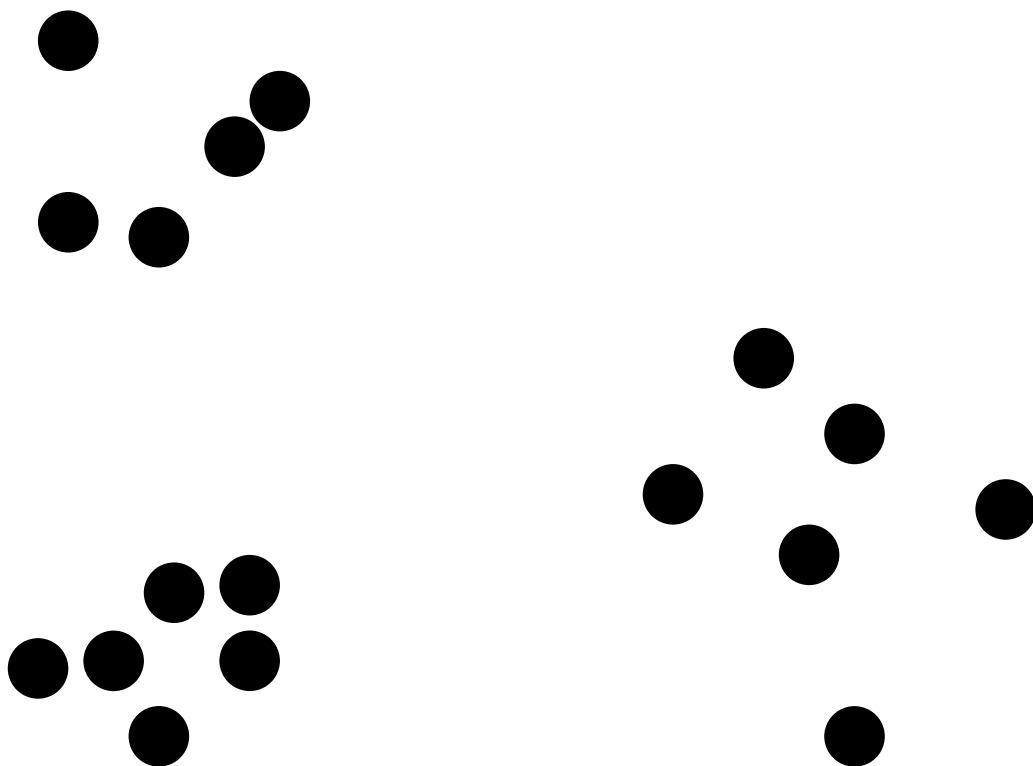
The supervised learner doesn't know much



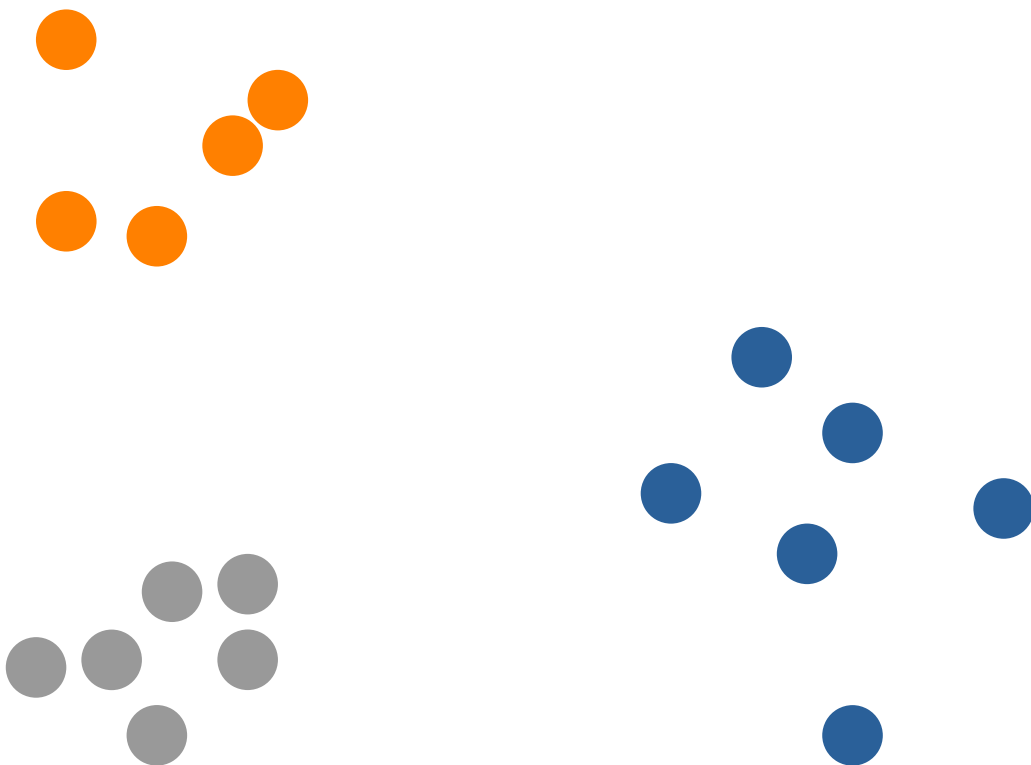
The unsupervised learner knows what it is doing



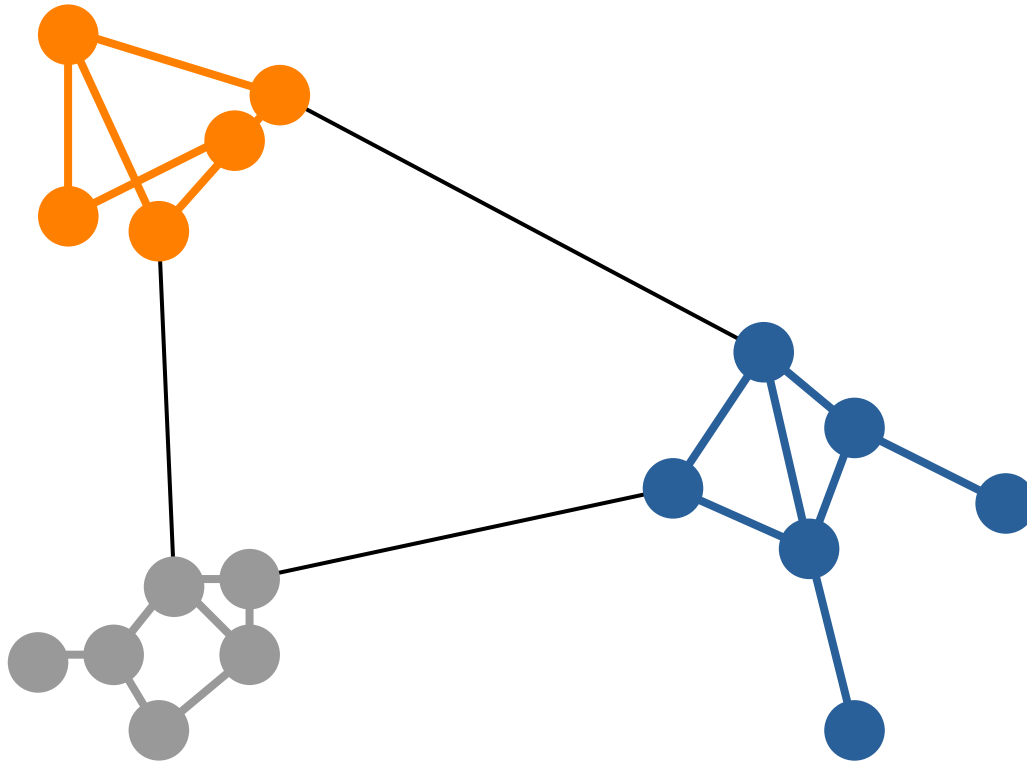
Clustering



Clustering



Community detection



"Cluster" these objects



"Cluster" these objects



Red

Not Red



"Cluster" these objects



Cannot Fly

Can Fly



"Cluster" these objects



Transport



Not Transport

"Cluster" these objects



Not Alive



Alive

There are no ground truth communities!

Modularity

A partition better than random chance

$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(x_i, x_j)$$

Modularity

A partition better than random chance

$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(x_i, x_j)$$



Value of the
adjacency matrix

Modularity

A partition better than random chance

$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(x_i, x_j)$$

Node degree

Value of the
adjacency matrix

Modularity

A partition better than random chance

$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(x_i, x_j)$$

Node degree

2 time the number of
edges in the network

Value of the
adjacency matrix

Modularity

A partition better than random chance

$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(x_i, x_j)$$

Node degree

2 time the number of
edges in the network

Value of the
adjacency matrix

Delta function: equals 1 if
nodes are in the same
community, or 0 if not.

Modularity

A partition better than random chance

$$Q = \sum_i (e_{ii} - a_i^2)$$

Modularity

A partition better than random chance

$$Q = \sum_i (e_{ii} - a_i^2)$$

Fraction of links
inside community i



Modularity

A partition better than random chance

$$Q = \sum_i (e_{ii} - a_i^2)$$

Fraction of links
inside community i



Expected fraction of
links in community i if
the network had no
community structure



Modularity

A partition better than random chance

$$a_i = \sum_j e_{ij}$$

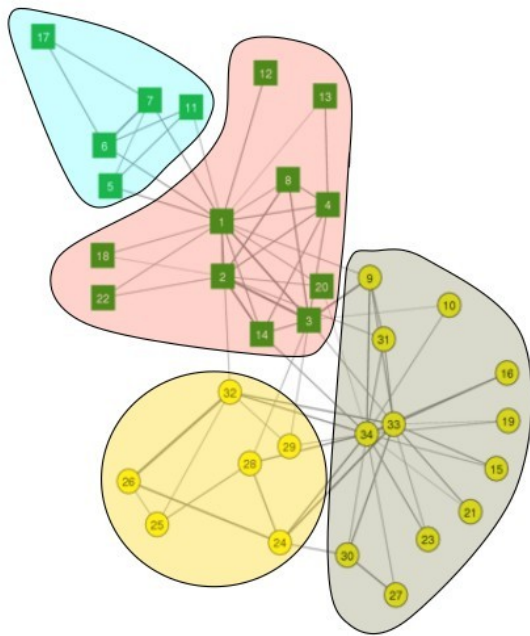
fraction of edges that
go to community i

$$Q = \sum_i (e_{ii} - a_i^2)$$

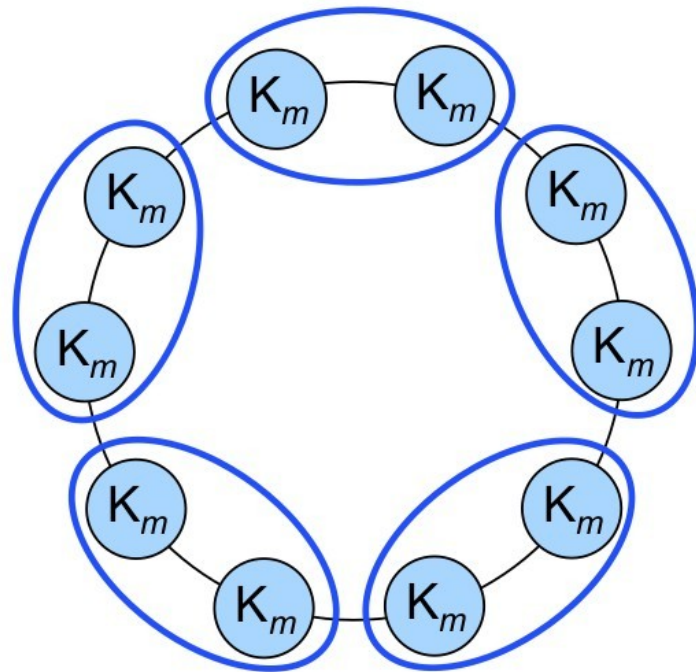
Fraction of links
inside community i

Expected fraction of
links in community i if
the network had no
community structure

Problems with modularity



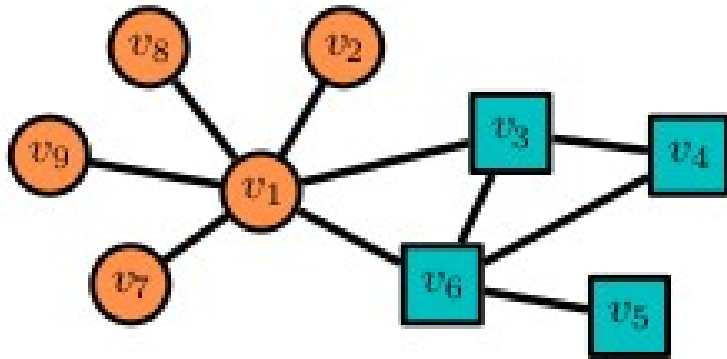
Finds spurious communities
(overfitting)



Resolution limit
(underfitting)

Calculate the modularity

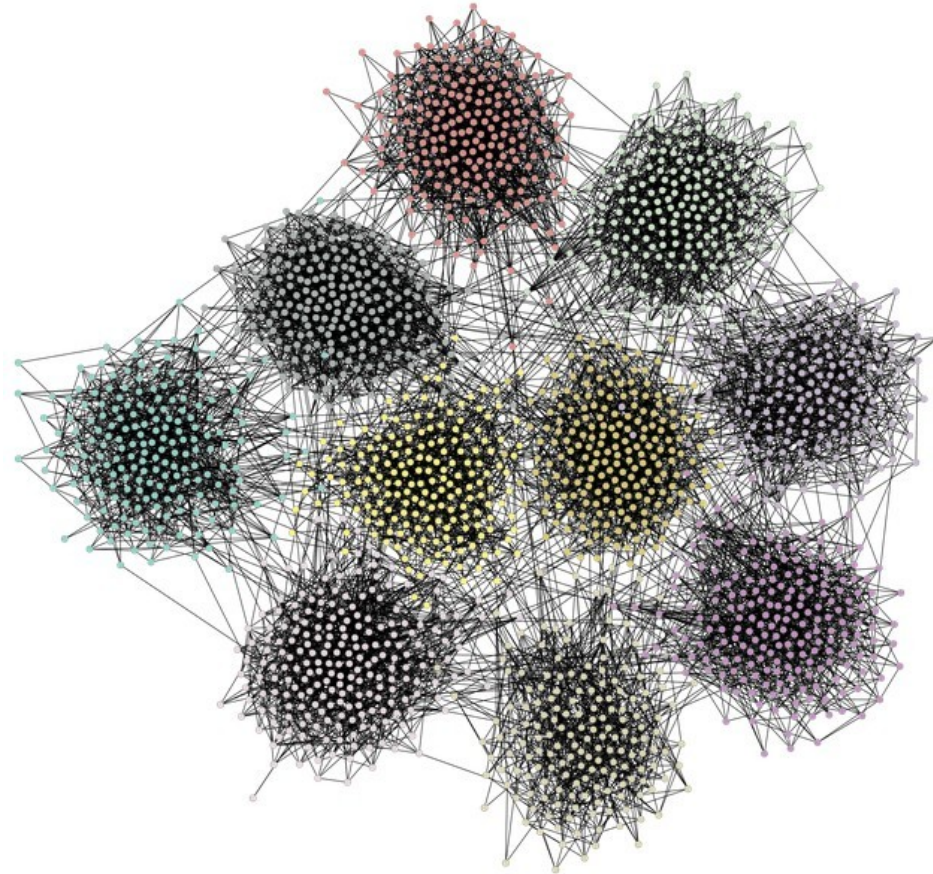
$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(x_i, x_j)$$



$$Q = \sum_i (e_{ii} - a_i^2)$$

Practical Q1 and Q2

Stochastic Block Models

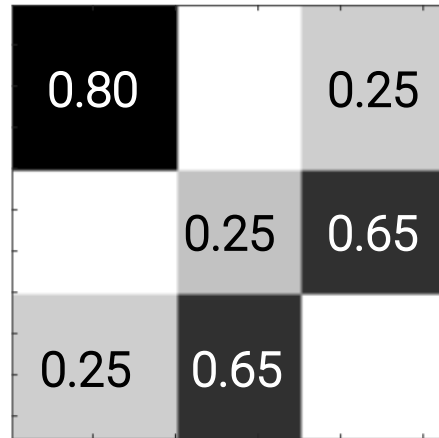


Generating a network using the SBM

Step 1 : Assign each node to a group

Generating a network using the SBM

Step 1 : Assign each node to a group



Mixing Matrix

- Step 2 : Select some connection probabilities (mixing matrix)

Generating a network using the SBM

Step 1 : Assign each node to a group

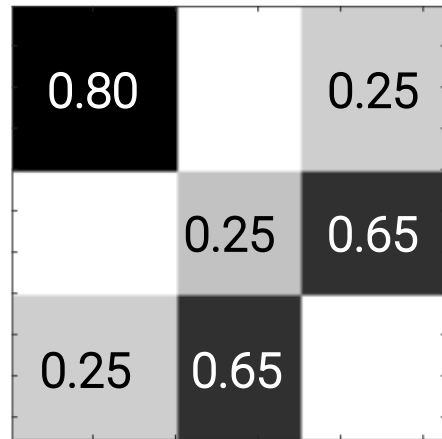


0.80		0.25
	0.25	0.65
0.25	0.65	

Mixing Matrix

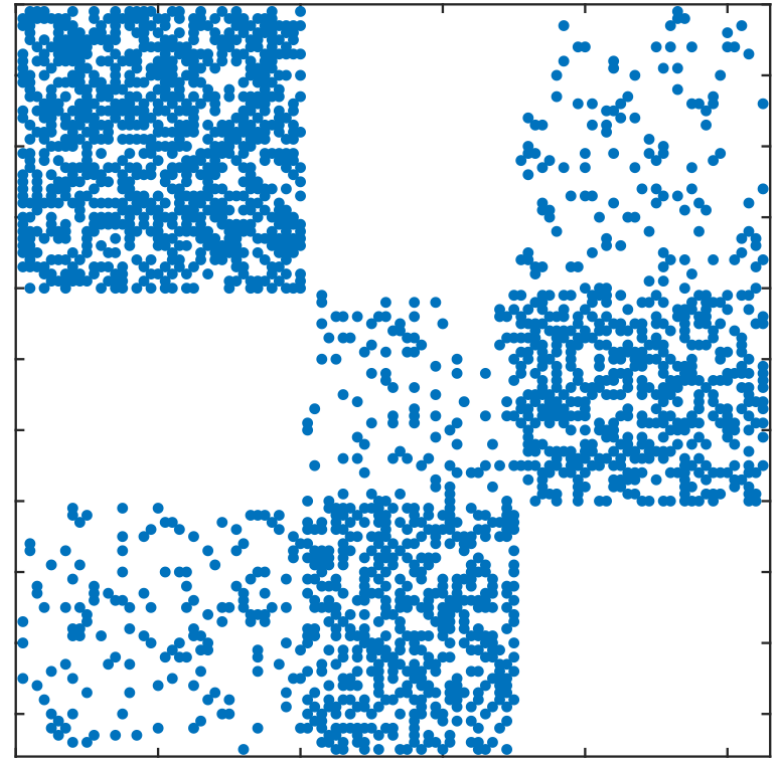
- Step 2 : Select some connection probabilities (mixing matrix)
- Step 3 : For each pair of nodes, add an edge with probability according to the group memberships

Generating a network using the SBM



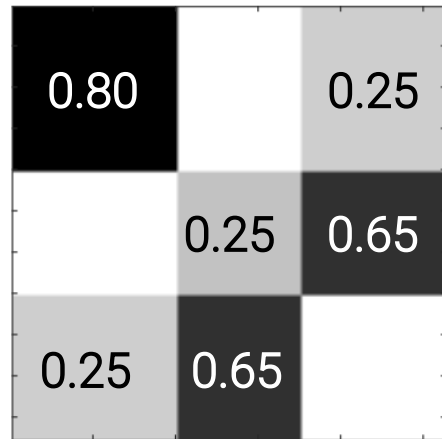
Mixing Matrix

generation
→



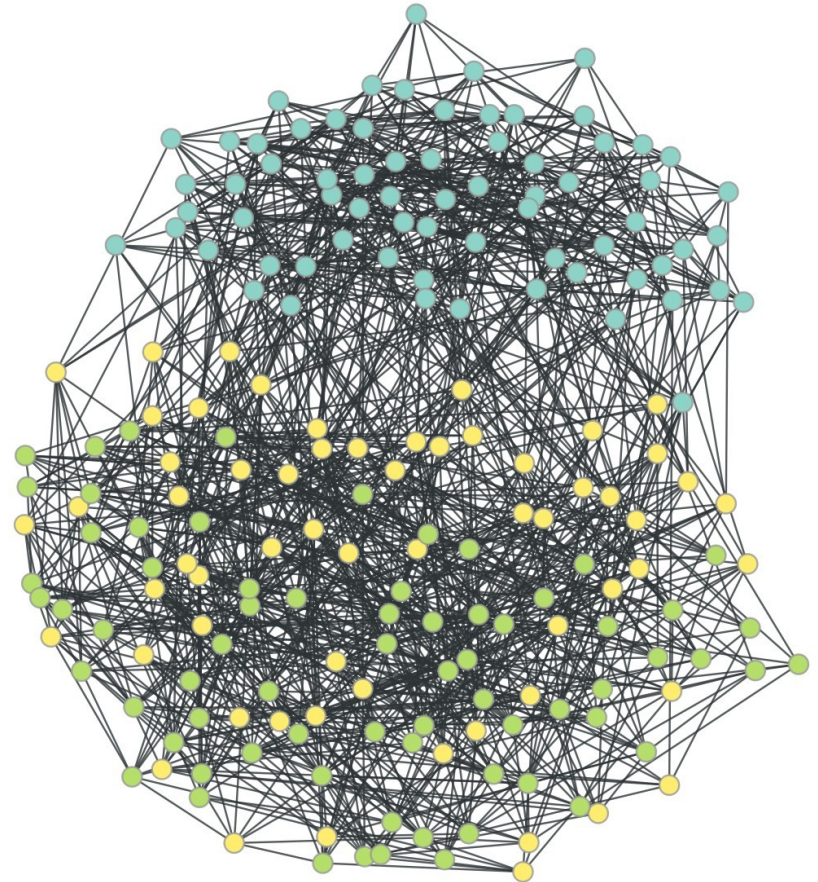
Adjacency Matrix

Generating a network using the SBM



Mixing Matrix

generation



Generating a network using the SBM

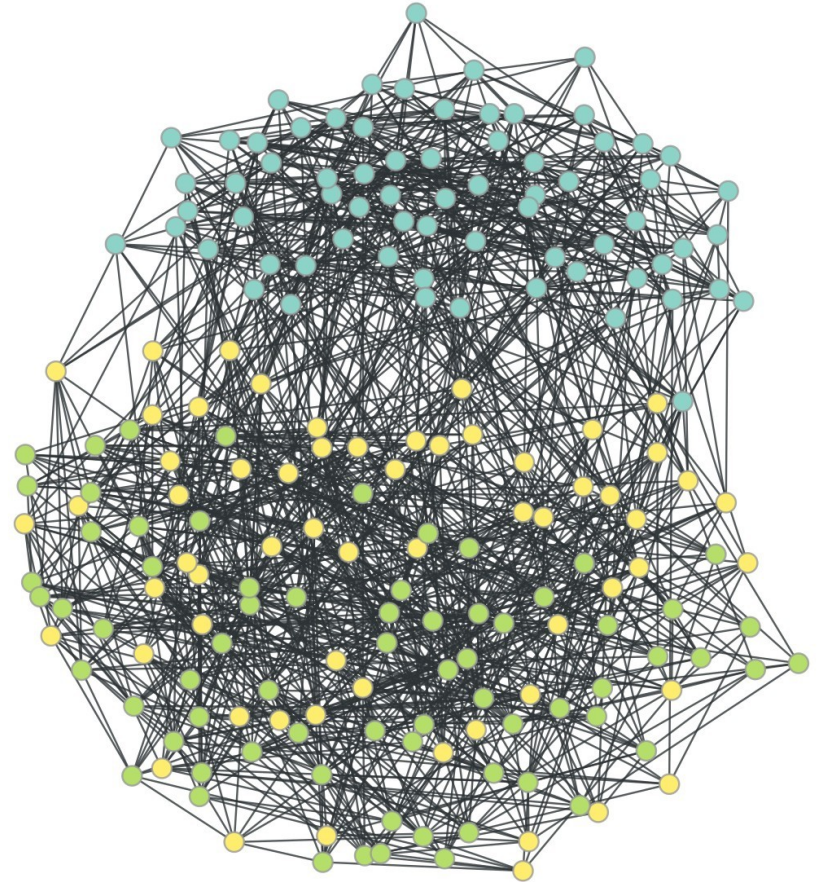
0.80		0.25
	0.25	0.65
0.25	0.65	

Mixing Matrix

generation

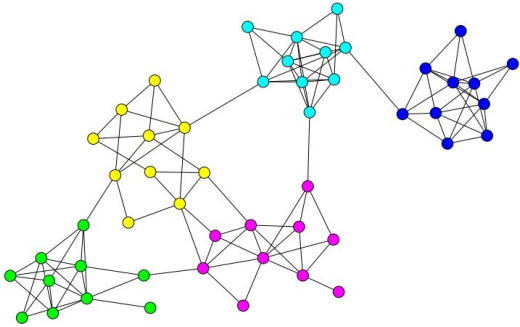
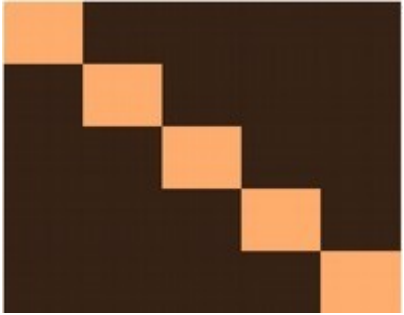


inference
(community
detection)



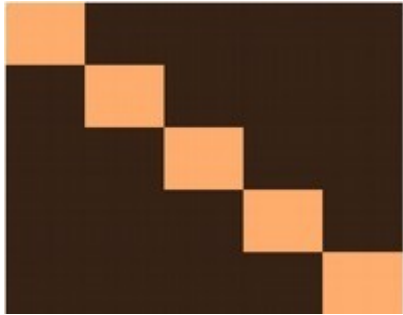
Different types of structure

assortative

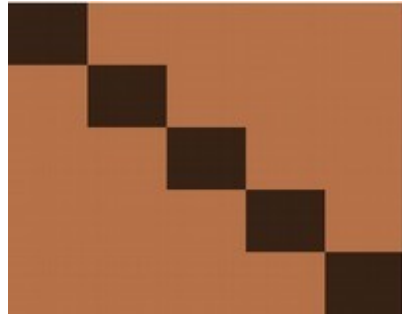


Different types of structure

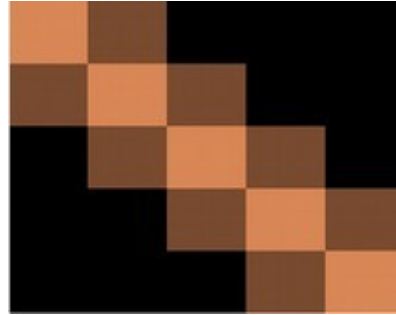
assortative



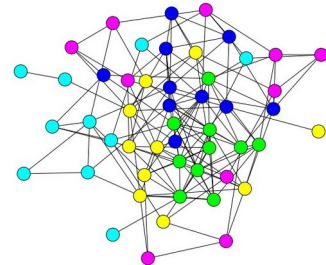
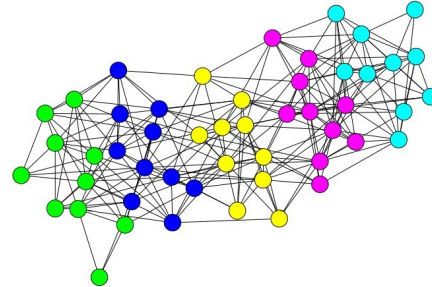
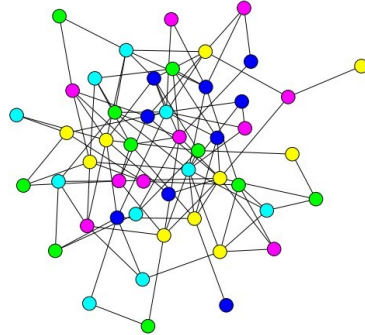
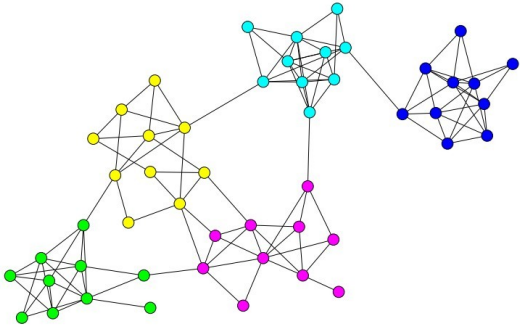
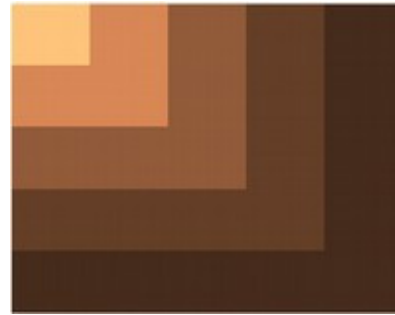
disassortative



ordered



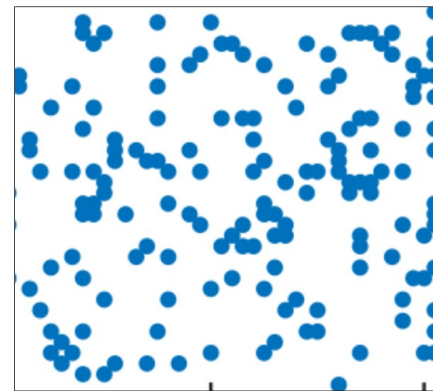
core-periphery



Erdos-Renyi random graphs

Flip biased coin for every node pair

$$\Pr(A|\theta) = \prod_{ij} \theta^{A_{ij}} (1 - \theta)^{(1-A_{ij})}$$



Adjacency Matrix

Heads or Tails?



Heads or Tails?



[0 1 0 1 0 1 0 0 1 0 0 0 0 0 1 0 0 0 1 1 1 1 1 1 1 0 1 1 0 0]

is just as likely as:

[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1]

What is the likelihood that we will observe this sequence of events?

$s = [0\ 1\ 0\ 1\ 0\ 1\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 0\ 1\ 1\ 0\ 0]$

What is the likelihood that we will observe this sequence of events?

$s = [0\ 1\ 0\ 1\ 0\ 1\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 0\ 1\ 1\ 0\ 0]$

$$\Pr(s_i = 1) = \theta = 0.5$$

$$(1-0.5) * 0.5 * (1-0.5) * 0.5 * (1-0.5) * 0.5 * (1-0.5) * (1-0.5) \dots$$

What is the likelihood that we will observe this sequence of events?

$s = [0\ 1\ 0\ 1\ 0\ 1\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 0\ 1\ 1\ 0\ 0]$

$$\Pr(s_i = 1) = \theta = 0.5$$

$(1-0.5) * 0.5 * (1-0.5) * 0.5 * (1-0.5) * 0.5 * (1-0.5) * (1-0.5) \dots$

$$\Pr(s|\theta) = \prod_i \theta^{s_i} (1 - \theta)^{(1-s_i)}$$

What is the probability of heads?

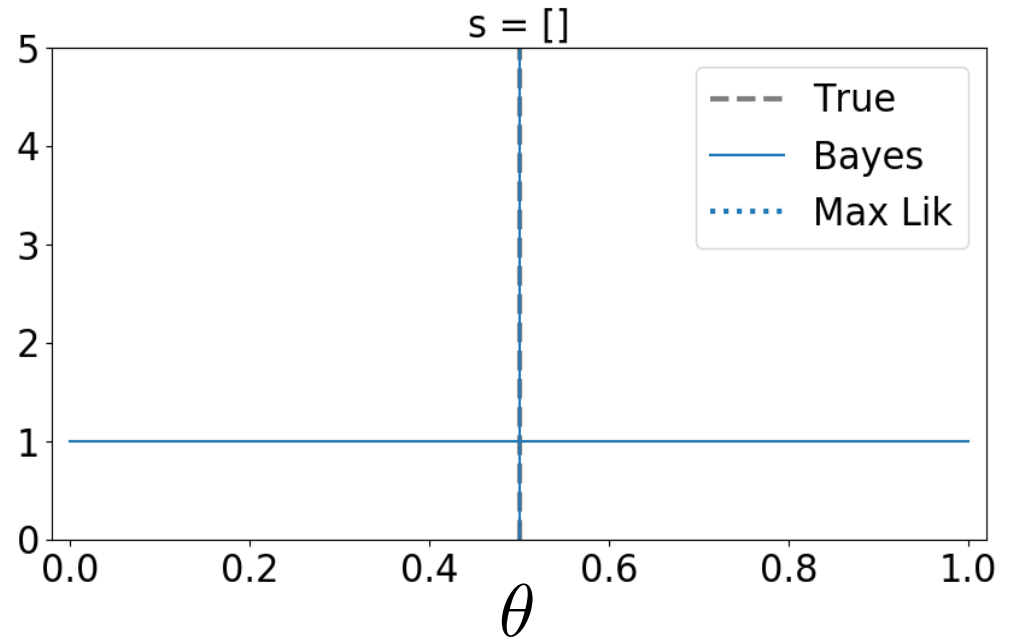


$$\theta_{\text{ML}} = \max_{\theta} \Pr(s|\theta)$$

$$\theta_{\text{Bayes}} \propto \Pr(s|\theta)\Pr(\theta)$$

Probabilistic generative model

What is the probability of heads?

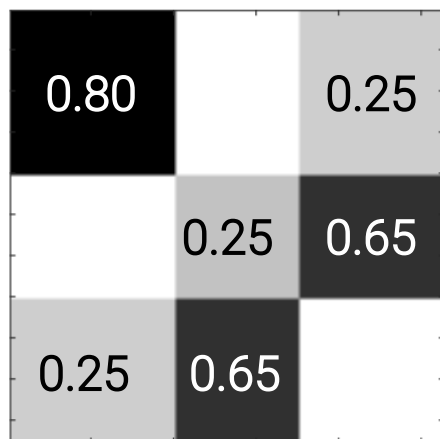


Probabilistic generative model

"I've never seen a coin before"

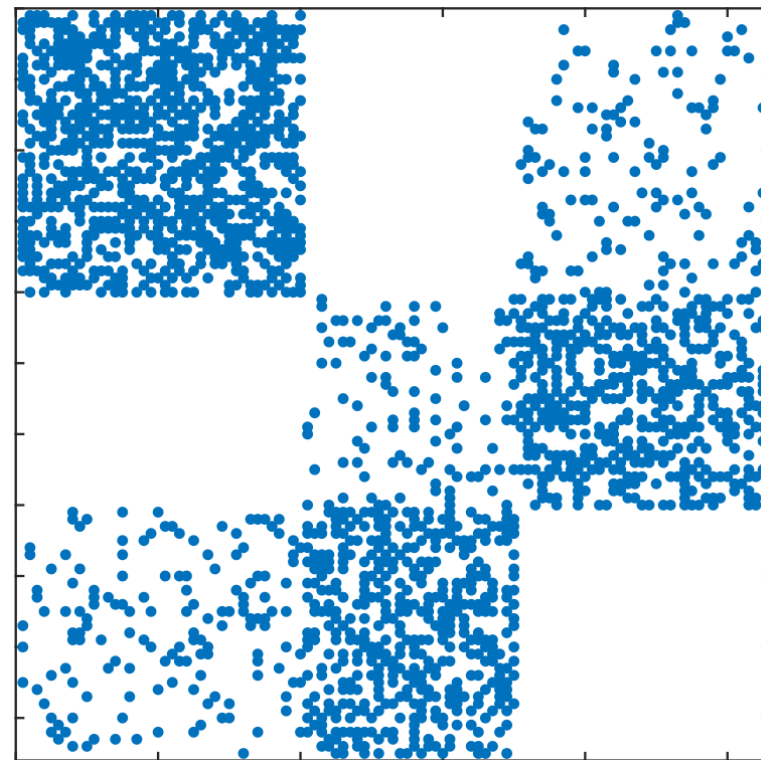
The stochastic block model

Just a bunch of coins!



Mixing Matrix

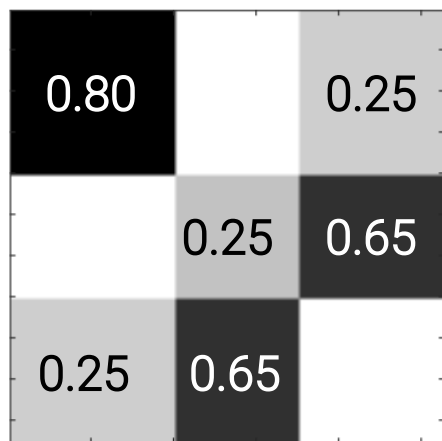
generation



Adjacency Matrix

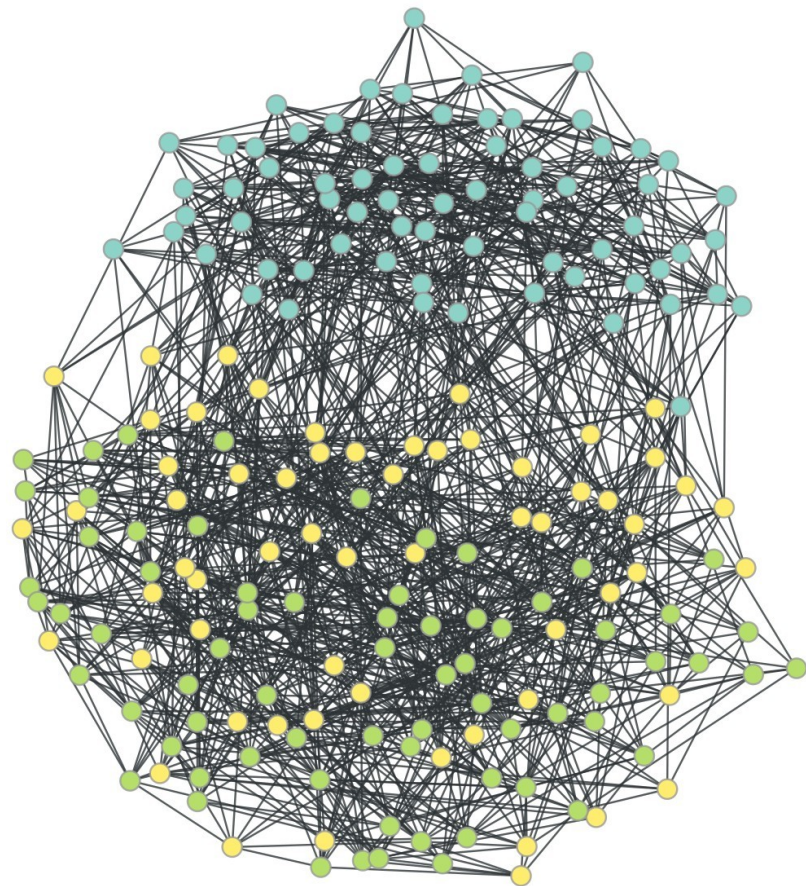
The stochastic block model

Just a bunch of coins!



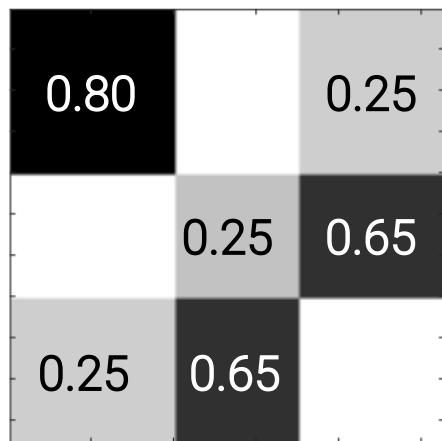
Mixing Matrix

generation



The stochastic block model

Just a bunch of coins!



Mixing Matrix

generation



inference



Minimum description length

Description length
(entropy)

$$\Sigma = -\ln P(\mathbf{A}|\boldsymbol{\theta}, \mathbf{b}) - \ln P(\boldsymbol{\theta}, \mathbf{b})$$

Minimum description length

Description length
(entropy)

$$\Sigma = - \ln P(\mathbf{A}|\boldsymbol{\theta}, \mathbf{b}) - \ln P(\boldsymbol{\theta}, \mathbf{b})$$

data description
length

Minimum description length

Description length
(entropy)

$$\Sigma = - \ln P(\mathbf{A}|\boldsymbol{\theta}, \mathbf{b}) - \ln P(\boldsymbol{\theta}, \mathbf{b})$$

data description
length

model description
length

Minimum description length

Description length
(entropy)

$$\Sigma = - \underbrace{\ln P(\mathbf{A}|\boldsymbol{\theta}, \mathbf{b})}_{\substack{\text{data description} \\ \text{length}}} - \underbrace{\ln P(\boldsymbol{\theta}, \mathbf{b})}_{\substack{\text{model description} \\ \text{length}}}$$

likelihood prior

The diagram illustrates the Minimum Description Length (MDL) principle. It features the equation $\Sigma = - \ln P(\mathbf{A}|\boldsymbol{\theta}, \mathbf{b}) - \ln P(\boldsymbol{\theta}, \mathbf{b})$. The first term, $-\ln P(\mathbf{A}|\boldsymbol{\theta}, \mathbf{b})$, is annotated with 'likelihood' above it and 'data description length' below it. The second term, $-\ln P(\boldsymbol{\theta}, \mathbf{b})$, is annotated with 'prior' above it and 'model description length' below it. To the left of the equation, the text 'Description length (entropy)' is present. Orange lines connect the labels to the corresponding parts of the equation.

Entropy is description length. Lower entropy, better model. No ground truth needed.

Practical Q3

Many extensions and
applications...

Link prediction

Network reconstruction

Many extensions

Many extensions and applications...

Link prediction

Network reconstruction

Many extensions

- degree correction
- mixed membership
- hierarchical
- edge weights/types
- node metadata
- temporal models

Many extensions and applications...

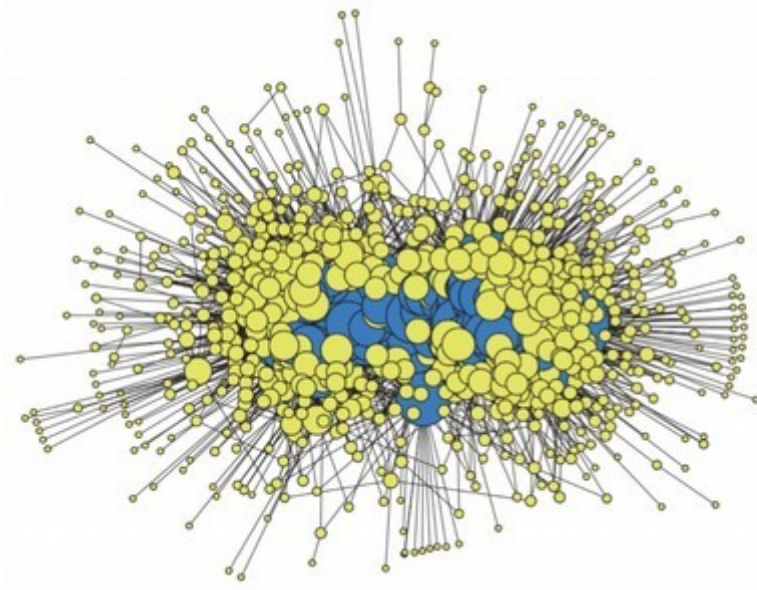
Link prediction

Network reconstruction

Many extensions

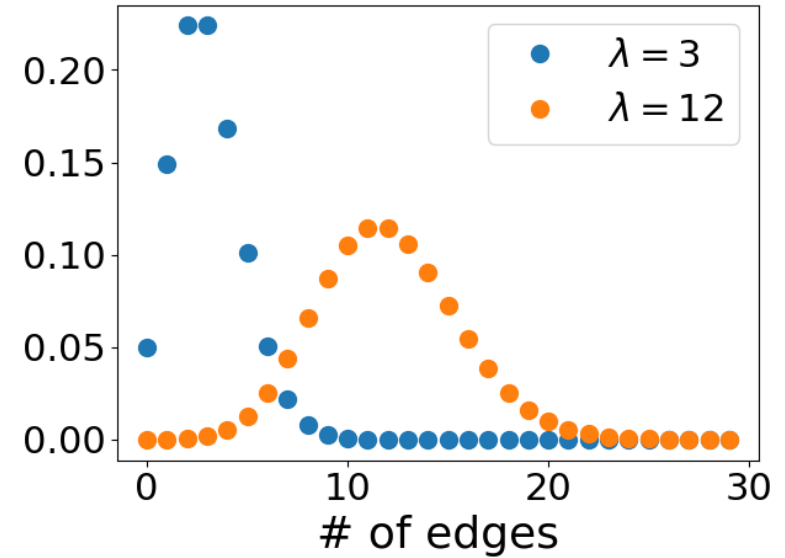
- degree correction
- mixed membership
- **hierarchical**
- edge weights/types
- node metadata
- temporal models

Degree-corrected SBM

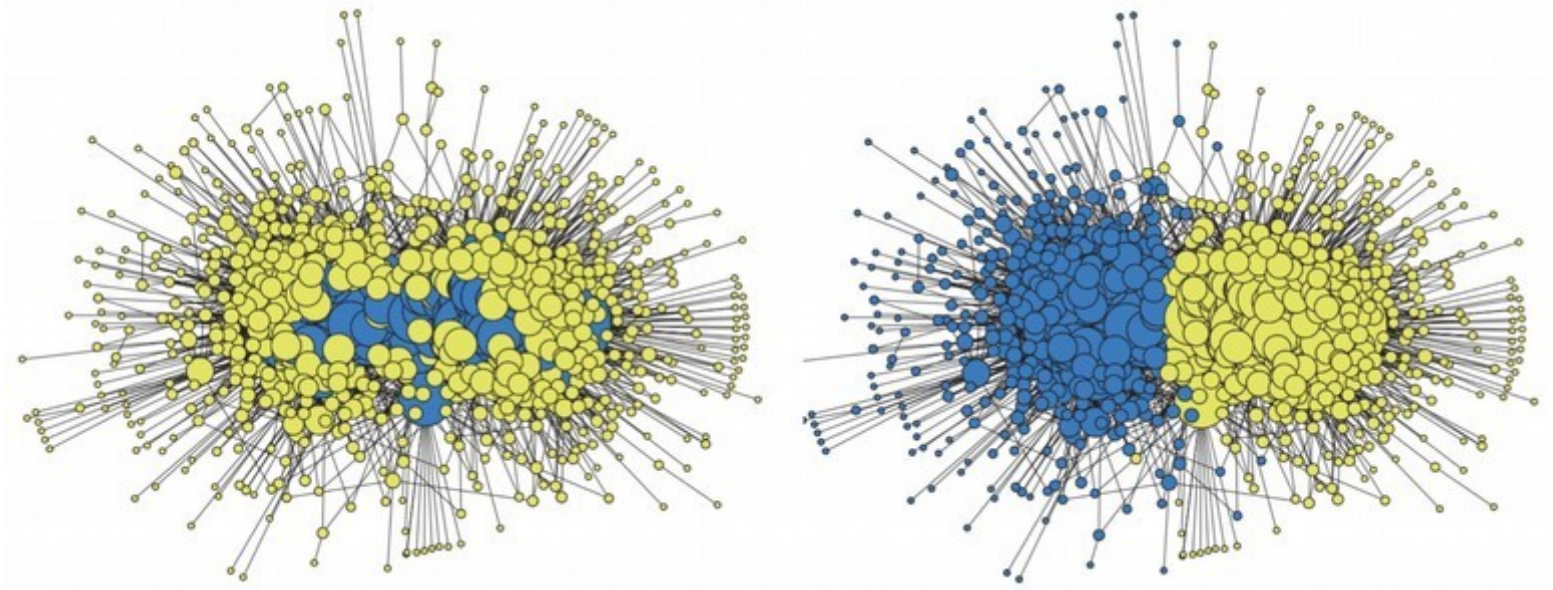


stochastic block model

SBM assumes Poisson distributed degree



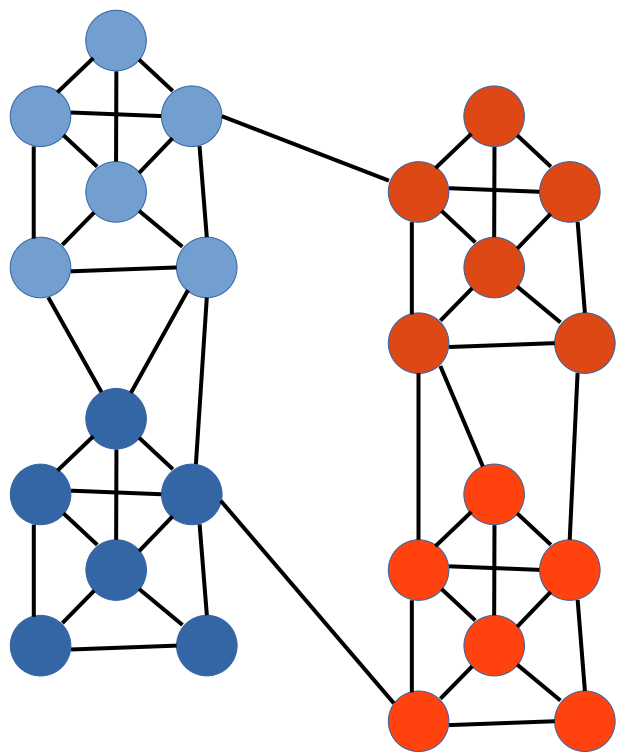
Degree-corrected SBM



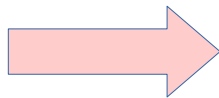
stochastic block model

Karrer, Newman. Stochastic blockmodels and community structure in networks. Phys. Rev. E 83, 016107 (2011).
Adamic, Glance. The political blogosphere and the 2004 US election: divided they blog. 36–43 (2005).

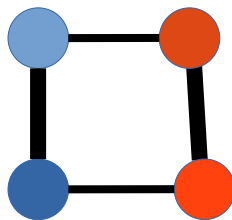
Building the hierarchy



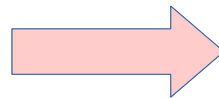
Observed network



infer
communities



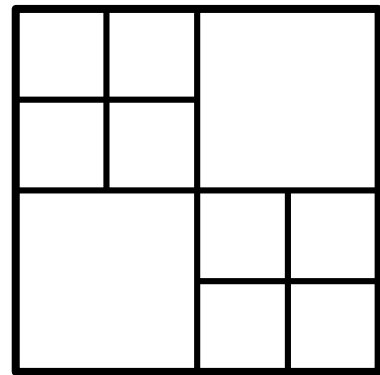
Multigraph

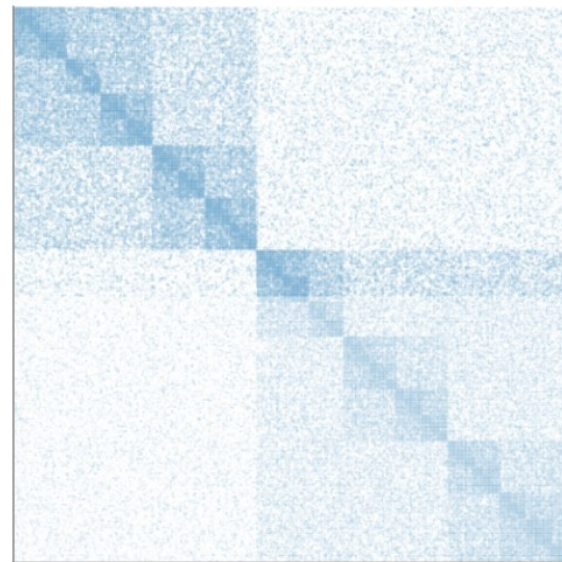
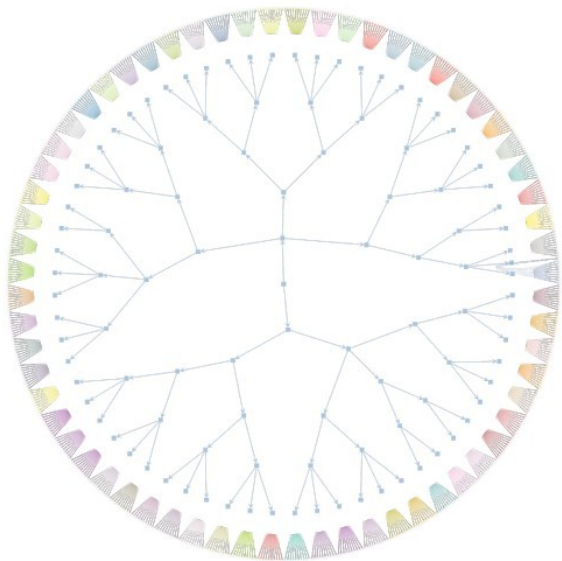


infer
communities



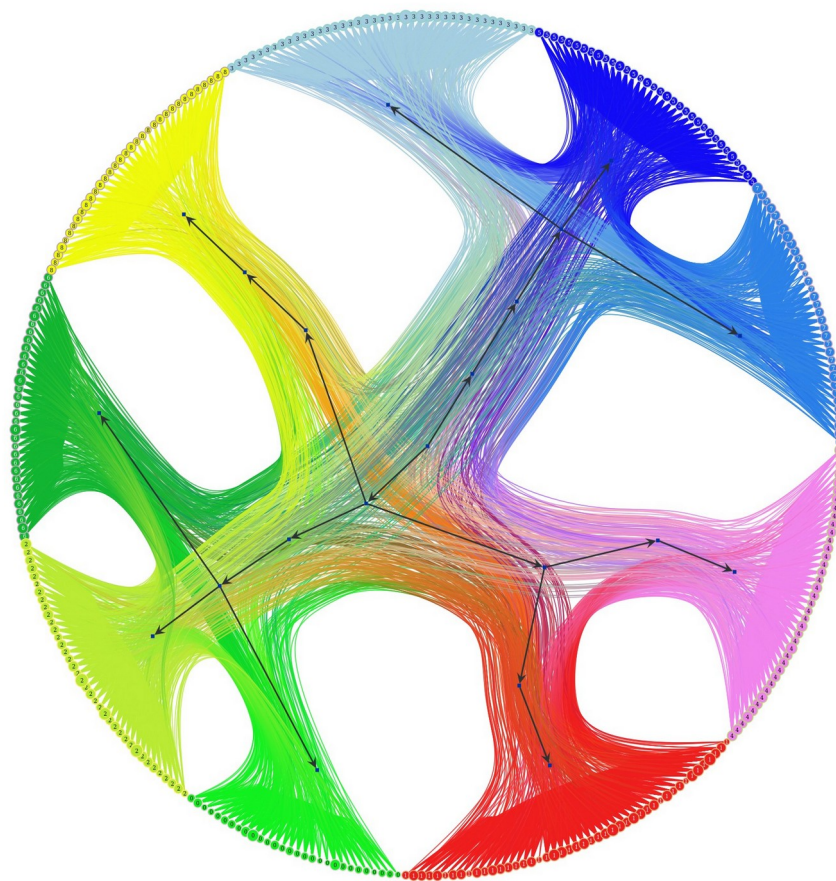
Multigraph



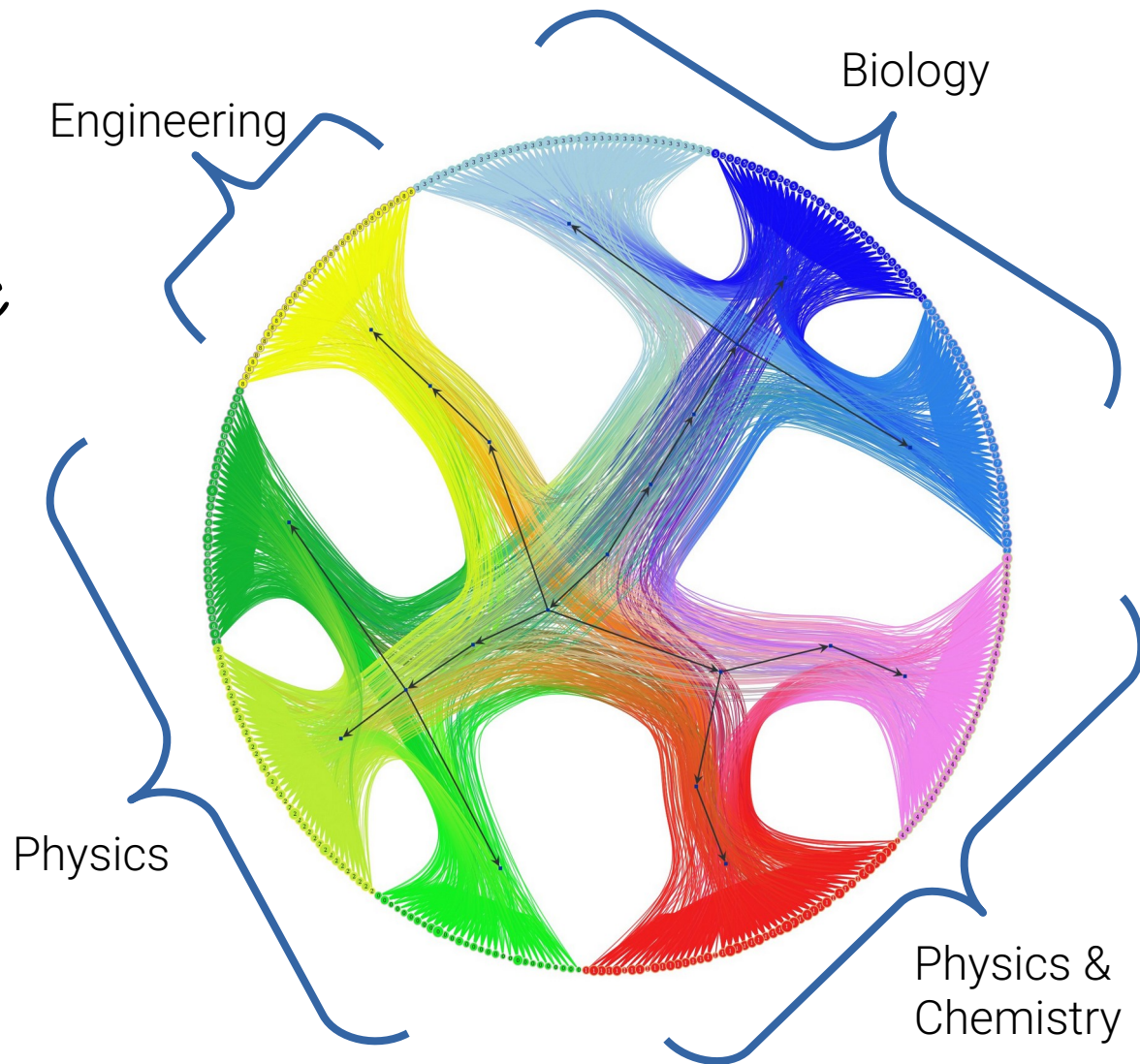


Adjacency matrix with hierarchy

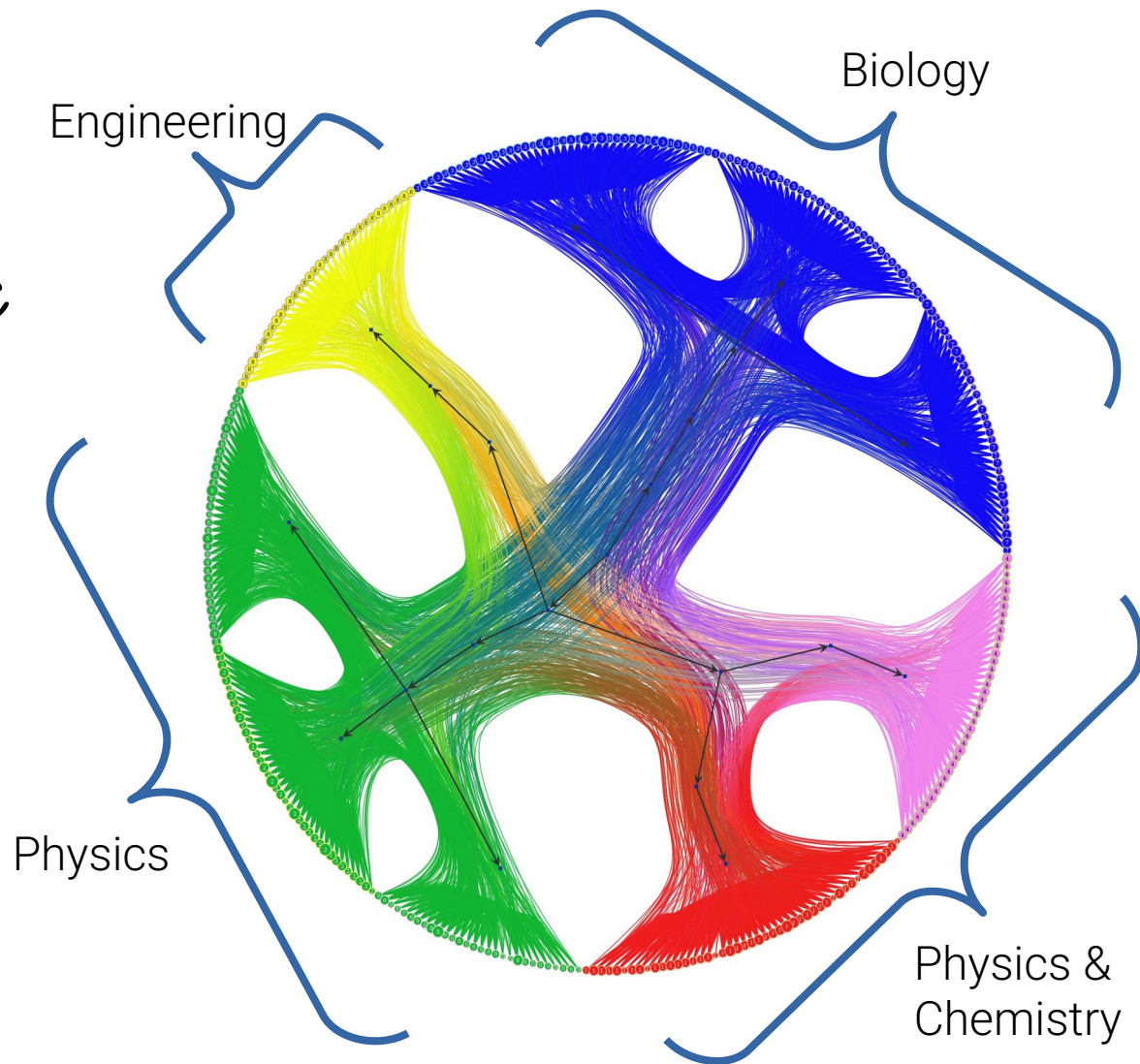
Face-to-face
contacts



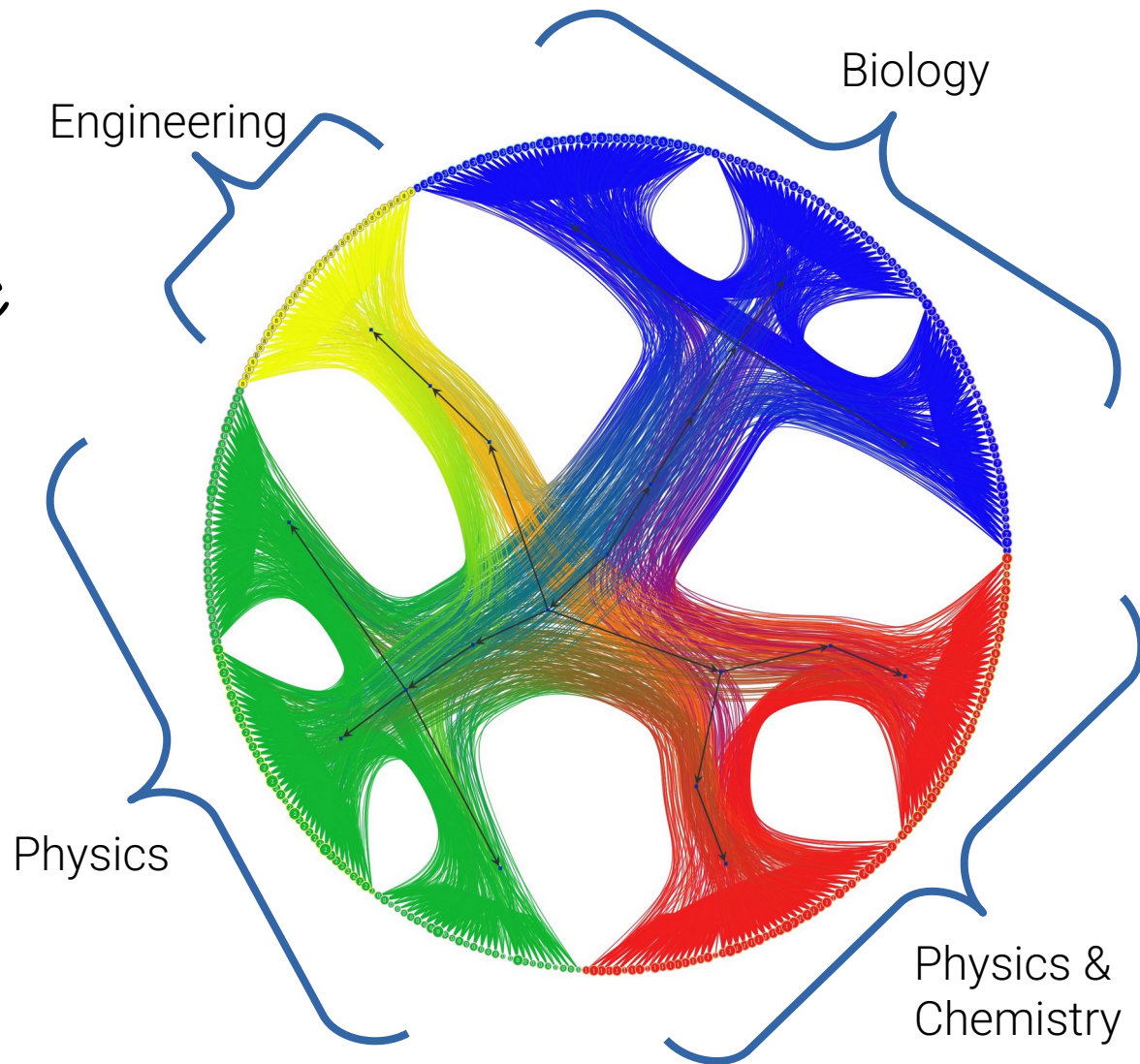
Face-to-face
contacts



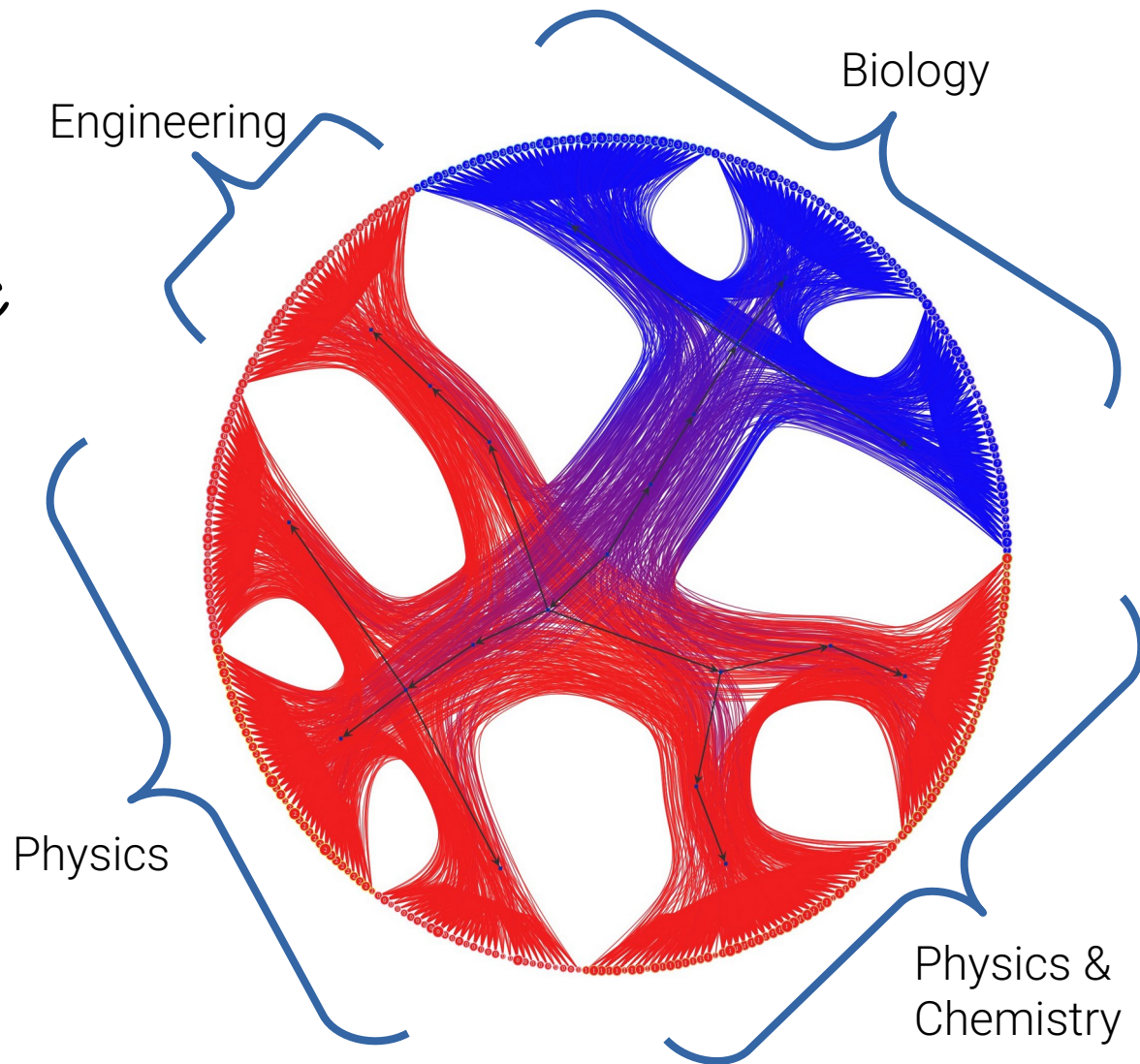
Face-to-face
contacts



Face-to-face
contacts



Face-to-face
contacts



Practical Q4 and Q5

Summary...

Summary...

Community detection finds nodes with similar connectivity patterns

- Nodes often have similar properties or functions

Summary...

Community detection finds nodes with similar connectivity patterns

- Nodes often have similar properties or functions

There can be multiple good ways to partition a network

Summary...

Community detection finds nodes with similar connectivity patterns

- Nodes often have similar properties or functions

There can be multiple good ways to partition a network

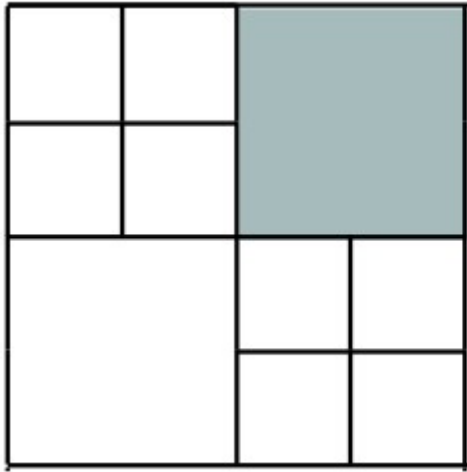
It's unsupervised!

- Your model must know how to partition the network

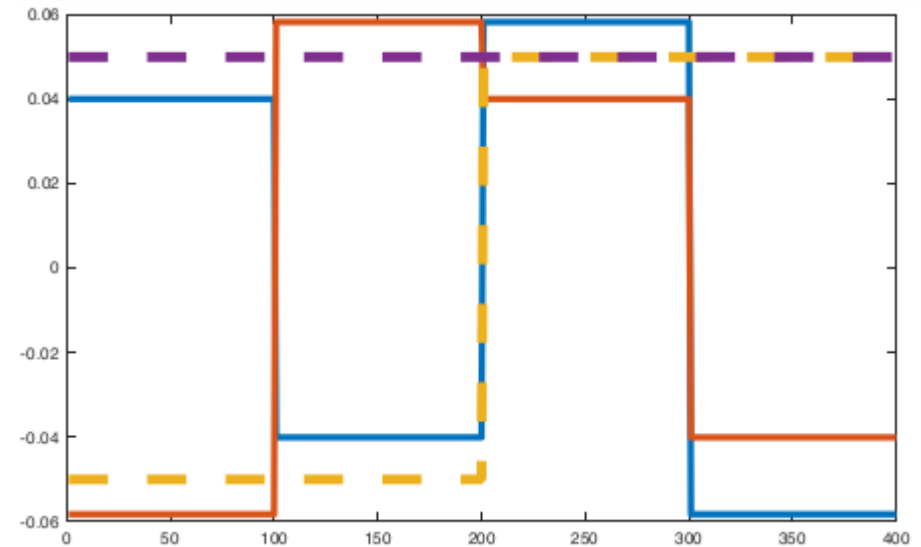
Spectral clustering

Spectral properties

$$\mathbb{E}[A]$$

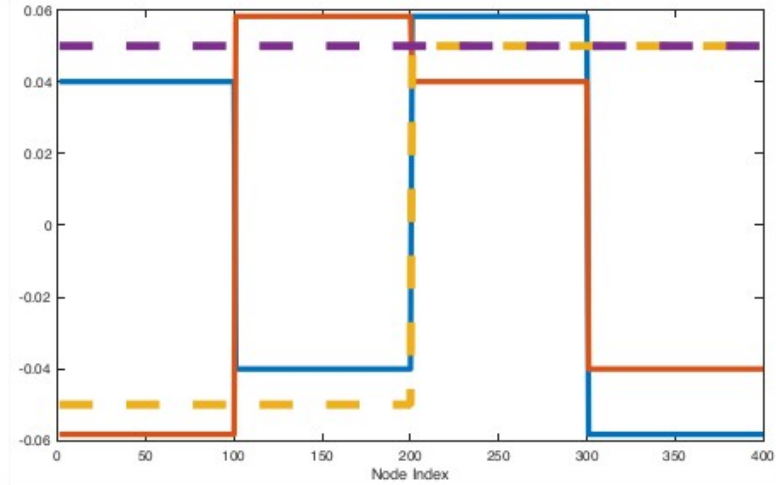


First 4 Eigenvectors of the Laplacian

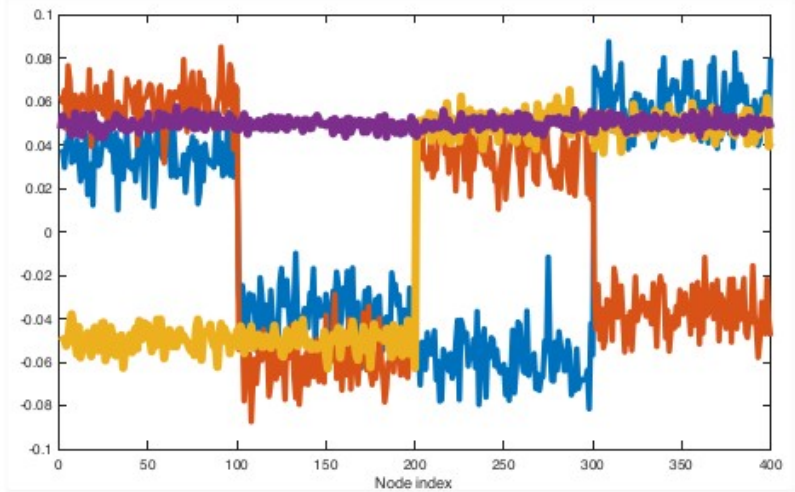
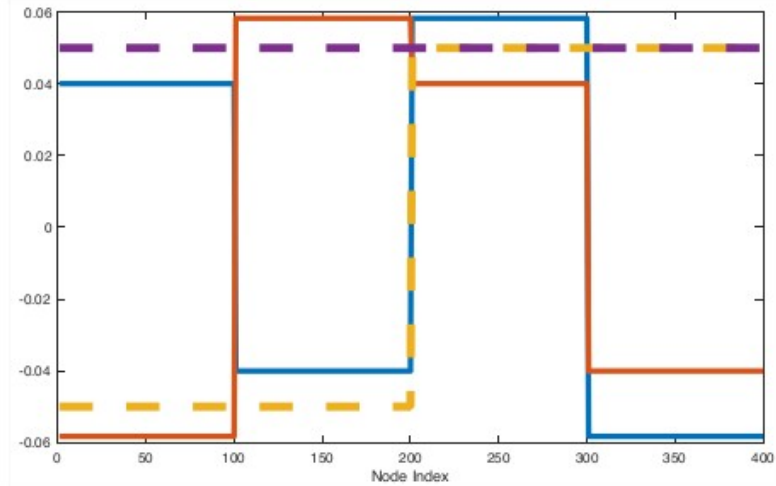


Node index

If we could just “see” the expected adjacency matrix,
then we could just look for constant eigenvectors



If we could just “see” the expected adjacency matrix,
then we could just look for constant eigenvectors



K-means clustering

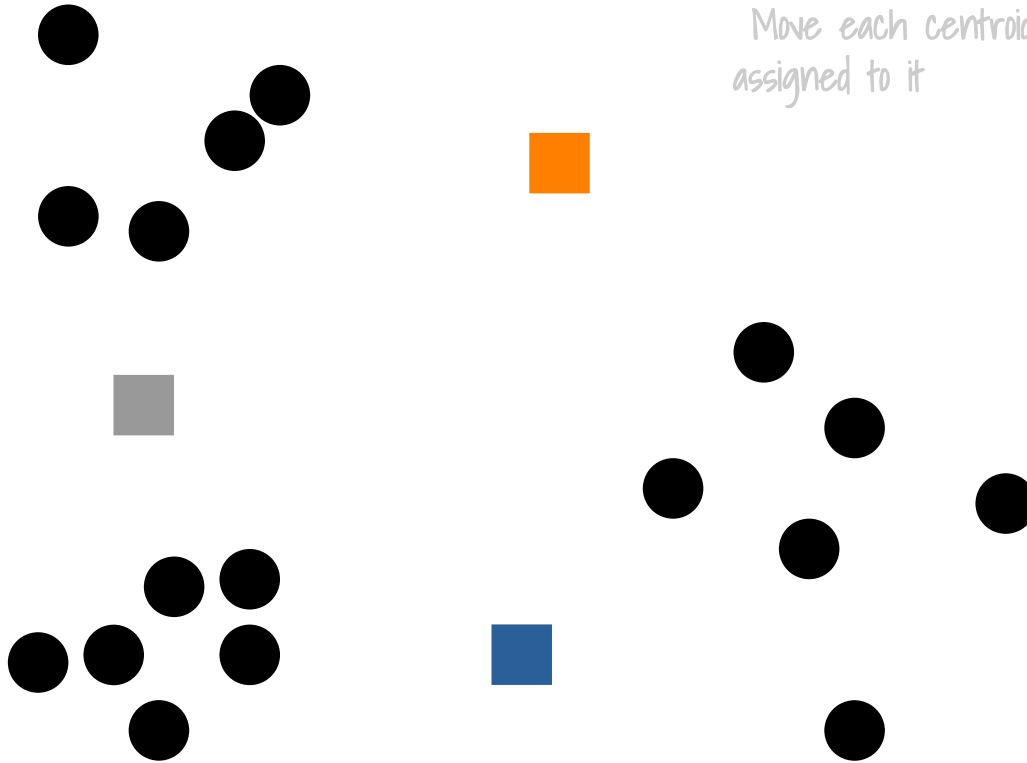
- 1) Randomly initialise centroids (one per cluster)
- 2) Iterate:
 - a) Assign each data point to the nearest centroid
 - b) Move each centroid to the mean of the data points assigned to it

Randomly initialise centroids (one per cluster)

Iterate:

Assign each data point to the nearest centroid

Move each centroid to the mean of the data points assigned to it

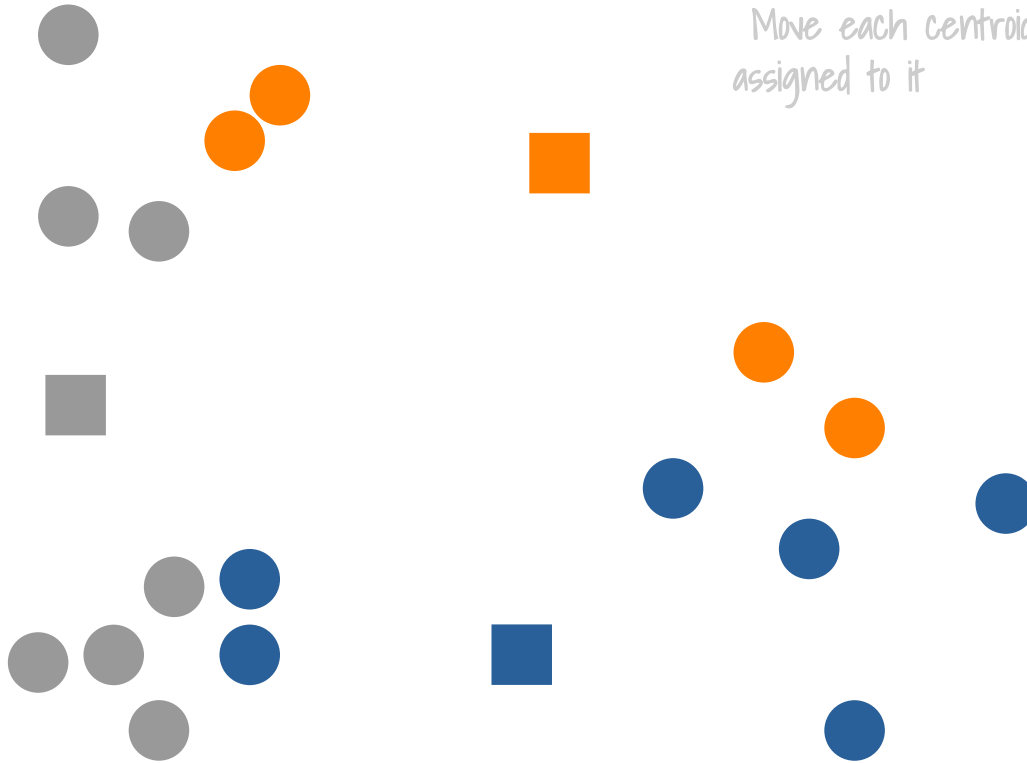


Randomly initialise centroids (one per cluster)

Iterate:

Assign each data point to the nearest centroid

Move each centroid to the mean of the data points assigned to it

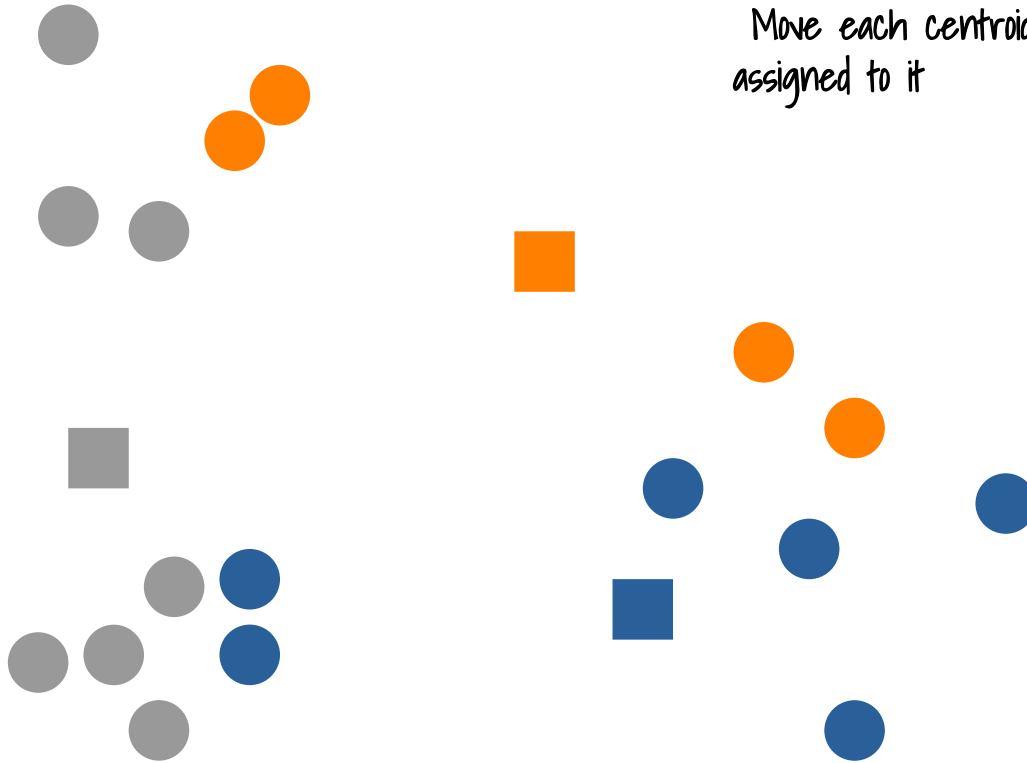


Randomly initialise centroids (one per cluster)

Iterate:

Assign each data point to the nearest centroid

Move each centroid to the mean of the data points assigned to it

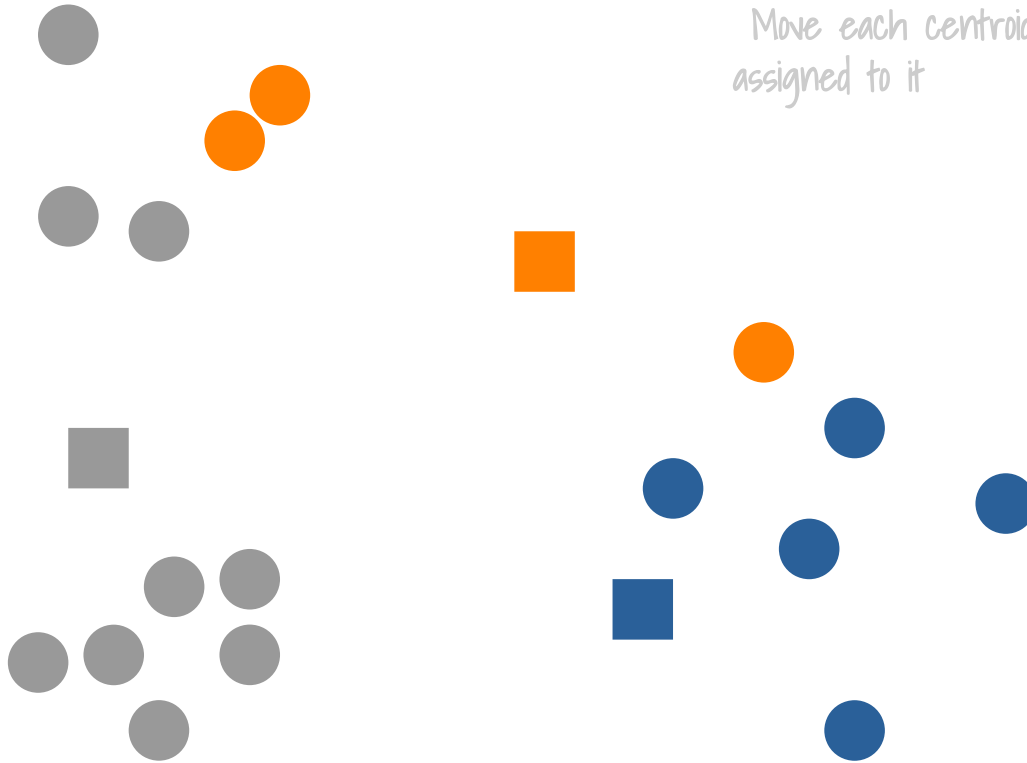


Randomly initialise centroids (one per cluster)

Iterate:

Assign each data point to the nearest centroid

Move each centroid to the mean of the data points assigned to it

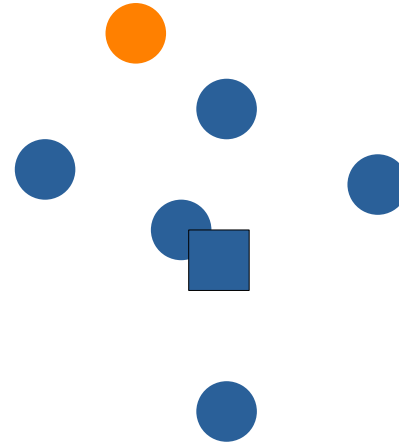
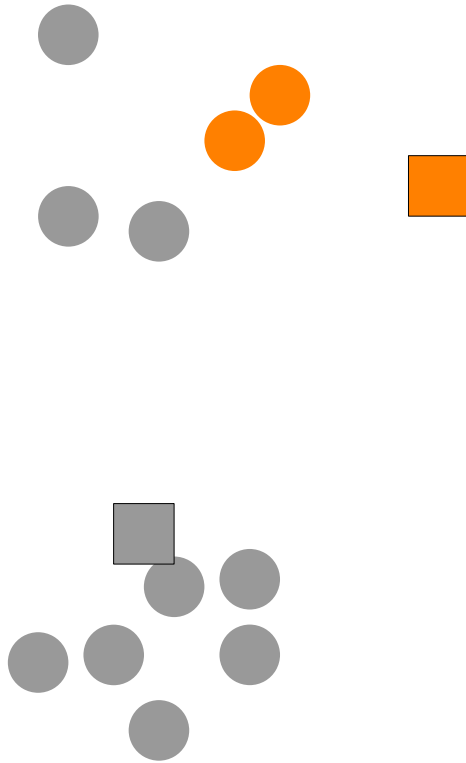


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Iterate:

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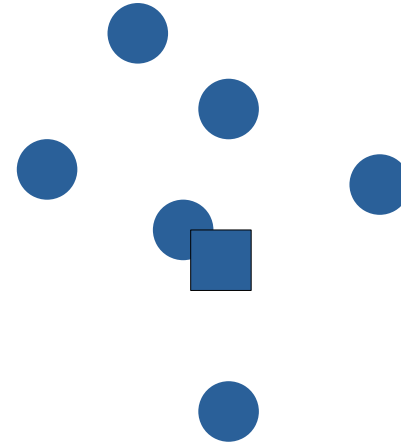
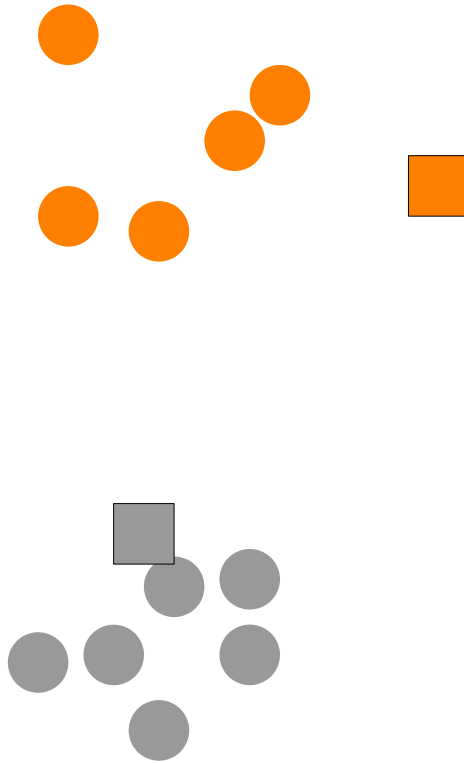


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